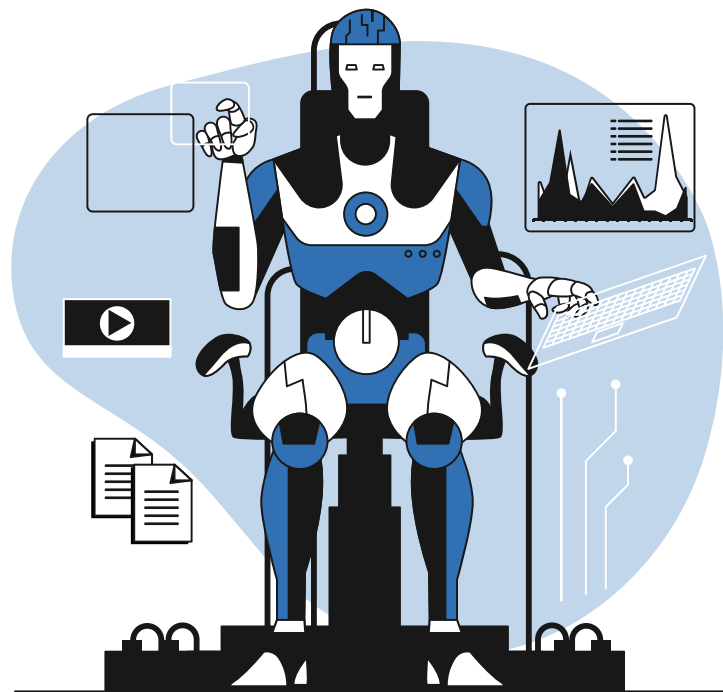


Real-Time Sound Source Localization and Tracking System

Samuele Ribaud, Hong Yan Jun



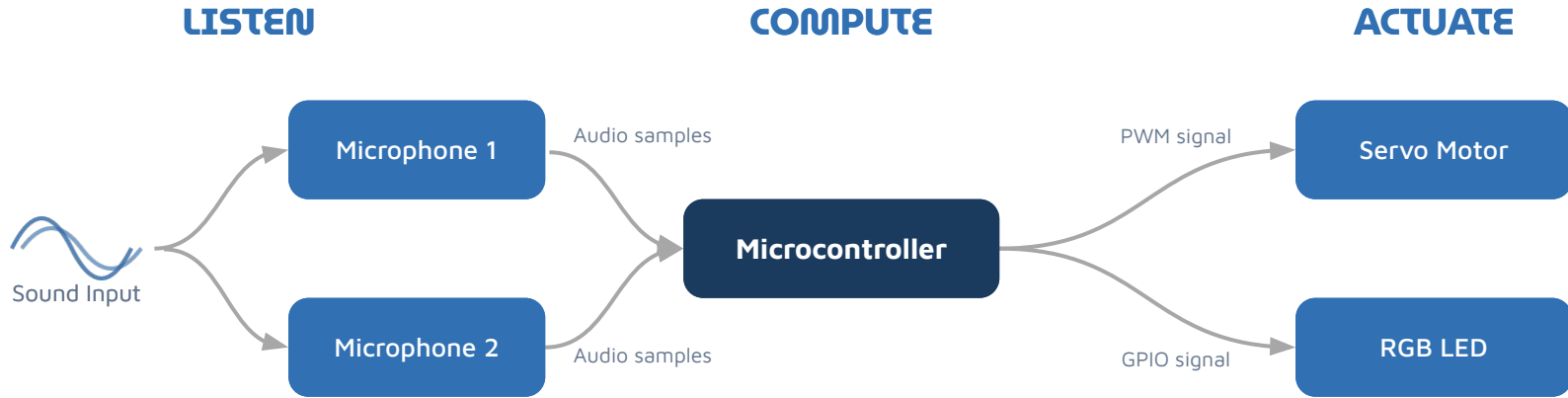


Project Concept

- Core Idea: Mimicking how human hear sounds
- Using an STM32 microcontroller and two standard microphones, the system acts like a pair of human ears.
- When a person speaks, the system calculates the sound's origin and uses a servo motor to physically rotate the sensor array toward the speaker.



System Workflow



Technical Details



- Eliminates the acoustic front-back 'cone of confusion' using a physical foam baffle to block rear audio signals.
- Demonstrates architectural depth by implementing the entire signal processing pipeline (filtering & cross-correlation) completely from scratch in standard C.
- "Employs a strict 'Listen-Compute-Move' finite state machine to prevent mechanical servo noise and vibrations from corrupting audio buffers.
- Converts the relative microphone time delays into an absolute coordinate space for precise 180° servo positioning.



Workload Distribution



Samuele:

1. Peripheral configuration (ADC, DMA, PWM, GPIO, Timers)
2. FSM and application logic
3. Angle computation
4. Actuation

Ryan:

1. CAD design & fabrication
2. Hardware assembly & wiring
3. DSP pipeline (manual)
 - Signal filtering
 - Cross-correlation & TDOA



Materials List



Material component Description	Qty	Target application / Notes	Sourcing / Cost
NUCLEO-U083RC Development Board	1	Main STM32 microcontroller unit	Lab inventory
MAX4466 analog microphones	2	Analog microphones for capturing audio	berrybase ↗ (€1,50 cad.)
MG996R 180° servo motor	1	Positional actuator for rotating the sensor platform	berrybase ↗ (€6,50)
Acoustic foam block	1	High-density foam sponge to block rear audio signals	Probably in the trash
RGB LED	1	4 pin LED for FSM state and limit reached indication	Lab inventory
PLA filament (3D Printing)	1	Material for printing the frame, ears, and servo mounts	Lab inventory
Assorted wires & breadboard	1	General circuit routing and power distribution	Lab inventory

